Algorithmic Fairness Definitions in Gans's Notation

Precision, Recall, Sufficiency, Separation, and Demographic Parity

ECON 470

Setup: Gans's Chapter 3 notation

We have a binary outcome $\theta \in \{0, 1\}$ (the true state — "positive" if $\theta = 1$, "negative" if $\theta = 0$) and a binary signal $s \in \{0, 1\}$ (the algorithm's prediction). The signal is characterized by two technological parameters:

$$e_1 = \Pr[s = 1 \mid \theta = 1] \quad \text{(true positive rate, sensitivity)}$$
$$e_2 = \Pr[s = 0 \mid \theta = 0] \quad \text{(true negative rate, specificity)}$$

Let $p = \Pr[\theta = 1]$ denote the *base rate* of positives in the population. For fairness analysis, we introduce a group indicator $G \in \{0, 1\}$ and allow (e_1, e_2, p) to differ across groups: write $e_1(g)$, $e_2(g)$, $p(g)$ for the group-conditional versions.

The 2×2 confusion matrix in Gans-language:

	$s = 1$	$s = 0$
$\theta = 1$	$e_1 \cdot p$	$(1 - e_1) \cdot p$
$\theta = 0$	$(1 - e_2)(1 - p)$	$e_2 \cdot (1 - p)$

(Cells give joint probabilities $\Pr[\theta, s]$.)

Part 1: Classification metrics

Precision $= \Pr[\theta = 1 \mid s = 1]$

The probability that a candidate flagged positive by the algorithm is actually positive — equivalently, the *posterior on* $\theta = 1$ *given* $s = 1$. By Bayes' rule:

$$\text{Precision} = \Pr[\theta = 1 \mid s = 1] \;=\; \frac{e_1\, p}{e_1\, p + (1 - e_2)(1 - p)}.$$

Intuition. Among everyone the algorithm labels as positive, what fraction really are positive? It depends on the **base rate** p as well as the signal quality (e_1, e_2).

False omission rate $= \Pr[\theta = 1 \mid s = 0]$

The probability that a candidate cleared by the algorithm is *actually* positive — the posterior on $\theta = 1$ given the negative signal $s = 0$. By Bayes' rule:

$$\Pr[\theta = 1 \mid s = 0] \;=\; \frac{(1 - e_1)\, p}{(1 - e_1)\, p + e_2(1 - p)}.$$

Intuition. Among everyone the algorithm clears, what fraction does it *miss* — i.e., is truly positive? The complement, $\Pr[\theta = 0 \mid s = 0]$, is the posterior that a cleared candidate really is negative. Like precision, both mix the base rate p with signal quality.

Recall $= \Pr[s = 1 \mid \theta = 1]$ **(sensitivity, true positive rate)**

This is e_1.

Intuition. Among everyone who is *truly* positive, what fraction does the algorithm catch?

Specificity $= \Pr[s = 0 \mid \theta = 0]$ **(true negative rate)**

This is e_2.

Intuition. Among everyone who is *truly* negative, what fraction does the algorithm correctly clear?

Part 2: Group-fairness criteria

Now allow $G \in \{0, 1\}$ to index two groups. Each criterion below requires a particular conditional probability to be equal across groups.

Independence (a.k.a. demographic parity, statistical parity)

Definition. The algorithm's prediction is independent of group membership:

$$s \perp G \qquad \Longleftrightarrow \qquad \Pr[s = 1 \mid G = 1] \;=\; \Pr[s = 1 \mid G = 0].$$

In Gans's notation:

$$e_1(1)\, p(1) + (1 - e_2(1))(1 - p(1)) \;=\; e_1(0)\, p(0) + (1 - e_2(0))(1 - p(0)).$$

Intuition. The algorithm flags the *same share* of each group as positive. If 30% of group 1 gets $s = 1$, then 30% of group 0 must also get $s = 1$.

Why many find it desirable. Demographic parity has the deepest roots in anti-discrimination law and practice of any of the three criteria. U.S. Title VII jurisprudence on **disparate impact** (Griggs v. Duke Power, 1971) treats a selection procedure that produces unequal pass rates across protected groups as presumptively unlawful — regardless of the procedure's accuracy or the employer's intent. The EEOC's "four-fifths rule" operationalizes this: if the selection rate for one group is less than 80% of the rate for another, the procedure is flagged for review. Demographic parity is the natural mathematical translation of that legal standard.

Three principled defenses are common:

1. **Outcomes, not predictions, are what people experience.** A person who is denied a loan, an interview, or pretrial release does not experience the algorithm's internal calibration; they experience the *decision*. If decisions land disproportionately on one group, that pattern is what shows up in lived outcomes, labor markets, neighborhood composition, and political representation. Demographic parity is the criterion that targets *that* — the distribution of allocations actually delivered.
2. **Base rates may themselves be artifacts of upstream injustice.** When $p(1) \neq p(0)$ in the training data, this rarely reflects a stable fact of nature. It can reflect biased policing, unequal access to education, historical lending discrimination, or other compounding inequities. An algorithm that treats those base rates as ground truth — and tunes prediction-side fairness criteria around them — bakes the upstream injustice into the downstream decision. Demographic parity is one way to refuse that.
3. **It is the only criterion that does not require ground-truth outcome data.** Separation and sufficiency both condition on θ — they need the actual outcome to be measured and recorded. In many real settings (hiring, lending, college admissions), outcomes for the *unselected* are never observed, and outcomes for the *selected* are observed under treatment, not control. Demographic parity sidesteps this by requiring only that selection rates be comparable.

Separation (a.k.a. equalized odds, equal opportunities)

Definition. The prediction is independent of group membership *conditional on the true outcome*:

$$s \perp G \mid \theta \qquad \Longleftrightarrow \qquad \Pr[s = 1 \mid \theta, G = 1] \;=\; \Pr[s = 1 \mid \theta, G = 0] \quad \text{for } \theta \in \{0, 1\}.$$

In Gans's notation, this is exactly the equality of e_1 and e_2 across groups:

$$\boxed{e_1(1) = e_1(0) \quad \text{and} \quad e_2(1) = e_2(0)}$$

Intuition. Conditional on being truly positive, both groups have the same probability of being flagged (e_1 equalized — Gans calls this "Equal Opportunities" on p. 335). Conditional on being truly negative, both groups have the same probability of being cleared (e_2 equalized; equivalently, false positive rates $1 - e_2$ are equalized — "Equal Odds" on p. 335).

Why many find it desirable. Separation is the criterion most closely aligned with what non-technical audiences usually mean by "the algorithm doesn't discriminate." It is the formal version of the principle that *qualified people should be treated the same regardless of group*. Three defenses are common:

1. **Equality of opportunity.** Equal e_1 across groups says: a qualified candidate ($\theta = 1$) from group 0 has exactly the same probability of being identified by the algorithm as a qualified candidate from group 1. This is the meritocratic intuition — "judged by the content of their character, not the color of their skin" — and it has bipartisan appeal across the political spectrum. It is the criterion Gans labels "Equal Opportunities" precisely because it matches the lay meaning of that phrase.
2. **The algorithm's mistakes should not be group-patterned.** When error rates differ across groups, the algorithm systematically fails some groups *as victims of error*. False

negatives in cancer screening that cluster by race; false positives in criminal-risk scoring that cluster by race — these are the harms that drove the COMPAS / ProPublica controversy and the broader public reaction to algorithmic decision-making. Separation directly targets that pattern: it says no group should bear a disproportionate share of the algorithm's mistakes.

3. **Procedural fairness.** Equal error rates are a form of *procedural* equal treatment, independent of distributive outcomes. The legal doctrine closest in spirit is disparate-*treatment* (as opposed to disparate-impact): the rule is that the *procedure* be the same for everyone, not that the outcomes equalize. Separation is the natural translation of "the procedure is the same" once you recognize that any noisy procedure makes errors — the procedure is the same iff the errors are distributed the same way conditional on truth.

Sufficiency (a.k.a. predictive parity, calibration)

Definition. The true outcome is independent of group membership *conditional on the prediction*:

$$\theta \perp G \mid s \qquad \Longleftrightarrow \qquad \Pr[\theta = 1 \mid s, G = 1] \;=\; \Pr[\theta = 1 \mid s, G = 0] \quad \text{for } s \in \{0, 1\}.$$

In Gans's notation, this says the **posterior** (precision when $s = 1$, and the false omission rate when $s = 0$) is the same in both groups:

$$\frac{e_1(1)\,p(1)}{e_1(1)\,p(1) + (1 - e_2(1))(1 - p(1))} \;=\; \frac{e_1(0)\,p(0)}{e_1(0)\,p(0) + (1 - e_2(0))(1 - p(0))},$$

and analogously for the negative side:

$$\frac{(1 - e_1(1))\,p(1)}{(1 - e_1(1))\,p(1) + e_2(1)(1 - p(1))} \;=\; \frac{(1 - e_1(0))\,p(0)}{(1 - e_1(0))\,p(0) + e_2(0)(1 - p(0))}.$$

Intuition. A risk score of "$s = 1$" means the *same thing* — the same probability of being truly positive — regardless of which group the candidate is in. Equivalently: the algorithm is **equally well-calibrated** on both groups.

Why many find it desirable. Sufficiency is the criterion most familiar to actuaries, doctors, and bankers — anyone who has to act on a risk score in the real world. The intuition is simple: **a risk score should mean what it says, regardless of who is being scored.** Three defenses are common:

1. **Honest communication of risk.** If "$s = 1$" carries a posterior probability of, say, 70% in group 1 but only 40% in group 0, then the score is *lying* to the decision-maker about one of the two groups. A doctor told "high risk of recurrence" is now getting different actual probabilities depending on the patient's group, even though the words on the chart are the same. Sufficiency rules this out: it requires that the prediction be *self-consistent* across groups in the strongest sense — same prediction, same underlying probability.

2. **Decision-makers can act on the score uniformly.** When sufficiency holds, a decision-maker who applies the same threshold rule to all candidates — "treat if $s = 1$, don't if $s = 0$" — is making *consistent* decisions, in the sense that the expected outcomes of those decisions are the same across groups. This is why sufficiency is also called *predictive parity*: the predictions are equally good as predictions, group by group. Insurance pricing, medical diagnosis, and credit underwriting all implicitly assume sufficiency — otherwise the actuarial tables would have to carry an asterisk explaining which group they apply to.

3. **It rules out one form of algorithmic deceit.** A failure of sufficiency means the algorithm is, in effect, hiding base-rate information in the group label. Two candidates with the same predicted score have systematically different chances of being truly positive depending on group — which is to say, the algorithm "knows" something about each candidate that the score is not transmitting. Sufficiency is the criterion that says the score is *all* the relevant information. This is what Gans on p. 335 means when he writes that a violation of predictive parity "might indicate that deliberate bias has been built into the algorithm somewhere."

Summary table

Criterion	**Probabilistic form**	**In Gans's (e_1, e_2, p) notation**
Independence	$s \perp G$	$\Pr[s = 1 \mid G = 1] = \Pr[s = 1 \mid G = 0]$
Separation	$s \perp G \mid \theta$	$e_1(1) = e_1(0)$ and $e_2(1) = e_2(0)$
Sufficiency	$\theta \perp G \mid s$	$\Pr[\theta = 1 \mid s, G = 1] = \Pr[\theta = 1 \mid s, G = 0]$ for $s \in \{0, 1\}$

Part 3: The impossibility result (Kleinberg, Mullainathan & Raghavan 2016; Chouldechova 2017)

Standing assumptions (carried over from Chapter 3 §3.1.2). The signal is **informative but imperfect**:

$$e_1 + e_2 > 1 \qquad \text{and} \qquad e_2 < 1.$$

The first assumption is what makes s a useful signal — it implies $\Pr[\theta = 1 \mid s = 1] > p$ and $\Pr[\theta = 1 \mid s = 0] < p$ (i.e., the signal moves the posterior in the right direction; cf. Ch. 3 §3.1.2). Together they imply $e_1 > 1 - e_2 > 0$, so both error rates are strictly between 0 and 1.

Theorem. Under the standing assumptions, **any two** of {independence, separation, sufficiency} imply $p(0) = p(1)$.

Contrapositively: if the base rates differ across groups, no two of the three criteria can hold simultaneously. The three are *pairwise* incompatible whenever the population is genuinely non-homogeneous.

Why the three pairs all fail. Each follows from a one-line algebra check on the 2×2 table.

Separation + sufficiency $\Rightarrow p(0) = p(1)$. If separation holds, e_1 and e_2 are the same in both groups. Then precision in group g is

$$\text{Precision}(g) = \frac{e_1\, p(g)}{e_1\, p(g) + (1 - e_2)(1 - p(g))}.$$

Differentiating with respect to p:

$$\frac{d\,\text{Precision}}{dp} = \frac{e_1(1 - e_2)}{\left[e_1\, p + (1 - e_2)(1 - p)\right]^2} \;>\; 0,$$

since $e_1 > 0$ and $1 - e_2 > 0$ by the standing assumptions. Precision is strictly increasing in the base rate: more of the population is genuinely positive, so a flagged candidate is more likely to be a true positive. Sufficiency — equal precision across groups — therefore forces $p(0) = p(1)$.

Separation + independence $\Rightarrow p(0) = p(1)$. If separation holds, the share flagged positive in group g is

$$\Pr[s = 1 \mid G = g] = e_1\, p(g) + (1 - e_2)(1 - p(g)) = (1 - e_2) + (e_1 + e_2 - 1)\, p(g).$$

Independence requires this to be the same in both groups, so $(e_1 + e_2 - 1)\,[p(1) - p(0)] = 0$. Since $e_1 + e_2 - 1 > 0$ by the standing assumption that the signal is informative, this forces $p(0) = p(1)$.

Sufficiency + independence $\Rightarrow p(0) = p(1)$. By the law of total probability, in each group:

$$p(g) = \underbrace{\Pr[\theta = 1 \mid s = 1, G = g]}_{\text{Precision}(g)} \cdot \Pr[s = 1 \mid G = g] \;+\; \underbrace{\Pr[\theta = 1 \mid s = 0, G = g]}_{\text{False Omission Rate}(g)} \cdot \Pr[s = 0 \mid G = g].$$

Sufficiency makes both posteriors the same across groups; independence makes both $\Pr[s = \cdot \mid G = g]$ terms the same across groups. So the entire right-hand side is the same in both groups, forcing $p(0) = p(1)$.

In each case, the only escape is equal base rates. Once $p(0) \neq p(1)$, the three criteria become pairwise mutually exclusive.

Part 4: After the impossibility — picking among the three

The impossibility result rules out having any two out of the three criteria at once, but it does not tell us which to give up. Each criterion has a real constituency, and the trade-off looks different depending on the application.

What each criterion costs you

Throughout this part, assume base rates genuinely differ ($p(0) \neq p(1)$) — the regime where the impossibility bites. By the theorem, enforcing any one criterion in that regime breaks the other two simultaneously.

If you enforce ...	...you keep	...you give up
Independence	Equal selection rates across groups; legal cover under disparate-impact doctrine; works without ground-truth θ	Both separation and sufficiency. Conditional error rates (e_1, e_2) must differ across groups, and calibration must break — the score's meaning shifts with group membership
Separation	Equal error rates conditional on truth (e_1, e_2 equalized); the algorithm's mistakes are not group-patterned	Both sufficiency and independence. A "high risk" score will mean different posterior probabilities of $\theta = 1$ depending on group (precision rises with the base rate); selection rates must differ across groups
Sufficiency	Calibration: a score means the same thing in both groups; what Gans calls "Predictive Parity"	Both separation and independence. The conditional error rates e_1 and e_2 must differ across groups, and the selection rate $\Pr[s = 1 \mid G]$ must differ as well

The case against each — stated honestly

Against independence. If base rates differ, demographic parity forces the algorithm to make systematically more errors on at least one group — it must either over-flag the lower-base-rate group or under-flag the higher-base-rate group. Demographic parity ignores θ entirely, so a procedure that satisfies it can be uniformly worse for *everyone* in both groups than a procedure that does not (the lemons-and-good-apples problem). Critics argue this fails the people in the disadvantaged group too: if the algorithm flags too many of them as positive in order to hit a parity target, individuals carry the cost of the population-level adjustment.

Against separation. Equalizing e_1 and e_2 across groups can require *miscalibration* — a "high risk" score is more reliable in one group than the other, and the decision-maker who acts on the score is no longer acting on what the score truthfully says. This can be costly when downstream decisions involve real stakes (releasing a defendant, extending a loan).

Against sufficiency. A well-calibrated algorithm can still produce wildly different selection rates and different error patterns across groups, leaving the lived-outcome disparities that independence targets fully intact. Calibration is a *predictive* virtue, not a *distributive* one — and in many policy contexts, distributive concerns are what motivated the audit in the first place.

Gans's §20.3 first-best as one way out

Rambachan, Kleinberg, Mullainathan & Ludwig (2020) — what Gans presents in §20.3 — proposes a different move entirely: **stop trying to make the prediction itself "fair," and instead separate the prediction step from the allocation step.**

- *Prediction step.* Train the algorithm to be as accurate as possible using *all* available information, **including group membership** G. This will, in general, satisfy none of the three criteria when base rates differ.

- *Allocation step.* Apply **group-specific thresholds** $t^*(g; K)$ to the resulting scores. The thresholds can encode whatever social-welfare or equity weights a planner wishes — and the choice of weights is where the value judgment honestly lives.

The appeal of this move is that it separates the **factual** question ("how likely is this candidate to repay the loan?") from the **normative** question ("how much should we weight outcomes in group 0 vs. group 1?"). Forcing both into a single fairness constraint on the prediction conflates them, and the impossibility result is what that conflation looks like mathematically.

This is one principled way out, not the only one — and a defender of independence (rooted in disparate-impact doctrine) might argue that legal and political legitimacy require constraints on selection rates regardless of what the planner's welfare function says. Knowing which trade-off you are making is the prerequisite for arguing about which to prefer.

Part 5: Quick lookup

Classification metric	**Probability**	**In** (e_1, e_2, p)
Precision	$\Pr[\theta = 1 \mid s = 1]$	$\frac{e_1 p}{e_1 p + (1-e_2)(1-p)}$
Recall / sensitivity	$\Pr[s = 1 \mid \theta = 1]$	e_1
Specificity	$\Pr[s = 0 \mid \theta = 0]$	e_2
False positive rate	$\Pr[s = 1 \mid \theta = 0]$	$1 - e_2$
False negative rate	$\Pr[s = 0 \mid \theta = 1]$	$1 - e_1$
Negative predictive value	$\Pr[\theta = 0 \mid s = 0]$	$\frac{e_2(1-p)}{e_2(1-p) + (1-e_1)p}$

Fairness criterion	**Equalize across groups**	**Gans label (p. 335)**
Independence	$\Pr[s = 1 \mid G]$	No Disparate Impact
Separation, $\theta = 1$ side	e_1	Equal Opportunities
Separation, $\theta = 0$ side	$1 - e_2$	Equal Odds
Sufficiency, $s = 1$ side	precision	Predictive Parity